

DTSA 5901

Special Topics

FORMAT	CATEGORY	CREDITS
Online (DTSA)	Elective	1
SECTION	BEST ROLE MATCH	AVG ALIGNMENT
Section 4 – Elective	Data Scientist (13.3 / 100)	8.8 / 100
SPECIALIZATION Special Topics		

9

CURRENT AVG SCORE
across 6 job roles

63

POTENTIAL SCORE
with recommended additions

+69

MAX SCORE GAIN
best single-role improvement

CURRENT COURSE DESCRIPTION

Examines a special topic in Data Science.

INDUSTRY ALIGNMENT BY JOB ROLE

JOB ROLE	CURRENT SCORE	PROJECTED SCORE	POINT GAIN	PRIORITY
Data Scientist	13.3	29.1	+15.8	MEDIUM
AI Engineer	10.9	66.8	+55.9	HIGH
Machine Learning Engineer	8.6	65.1	+56.5	HIGH
MLOps Engineer	7.8	69.6	+61.9	HIGH
ML Engineer	7.5	76.4	+68.9	HIGH
Applied Scientist	4.5	71.5	+67.0	HIGH

RECOMMENDED TOPICS TO ADD

Keywords highlighted in gold (**example**) appear as gaps across 3 or more job roles.

✓ Course vocabulary is well-aligned with industry job descriptions.

PRIORITY RECOMMENDATIONS FOR INSTRUCTORS



This course is already well-aligned with industry expectations. Continue monitoring for emerging tools and frameworks.

GAP KEYWORDS BY ROLE

Data Scientist

13.3 → 29.1 (+15.8 pts)

presentation

AI Engineer

10.9 → 66.8 (+55.9 pts)

models learning new tools make

Machine Learning Engineer

8.6 → 65.1 (+56.5 pts)

learning tools development models applications analysis statistical

MLOps Engineer

7.8 → 69.6 (+61.9 pts)

scientists learning media tools models development error

ML Engineer

7.5 → 76.4 (+68.9 pts)

media learning models world new tools

Applied Scientist

4.5 → 71.5 (+67.0 pts)

computational models department learning industry development fundamental world

Methodology: Scores (0–100) reflect TF-IDF cosine similarity between course catalog descriptions and a corpus of 100 LinkedIn data science job postings (collected June 2026, filtered to postings with ≥200 applicants). Gap keywords are terms that appear frequently in job postings for a given role but are absent or underrepresented in the current course description. "Projected score" shows the estimated alignment after integrating the top 5 suggested keywords. Analysis was conducted by the CU Boulder MSDS Curriculum-Industry Alignment project team.